

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Low Acrylonitrile NBR

1. Abstract

The patent landscape for Low Acrylonitrile NBR reveals a diverse range of synthesis and polymerization approaches. Patents EP2492306, US10266681, EP2463335, and EP3231826 demonstrate various methods for producing cross-linkable nitrile rubber compositions and graft copolymers. Recurring variables include temperature, viscosity, and conversion rates, while trends indicate a focus on optimizing synthesis conditions to achieve desired material properties. Notably, the use of specific initiators, emulsifiers, and cross-linking agents is prevalent across the patents. However, the specific acrylonitrile content is only disclosed in patent EP2463335, highlighting a gap in the available data.

2. Research Scope & Selection Method

This report includes patents that demonstrate direct synthesis or polymerization of Low Acrylonitrile NBR. The selection method involved identifying patents with relevant examples or procedures that detail the production of nitrile rubber compositions or graft copolymers. Only patents with explicit synthesis or polymerization methods were included, while those lacking specific experimental details were excluded.

3. Patent-by-Patent Analysis

PATENT EP2492306

Example EXAMPLE 1

Parameter	Value	Unit	Context
Brabender	55	-	Synthesis condition
Viscosity	100	-	Material property
Temperature	128	-	Synthesis condition

Rol	50	-	Synthesis condition
-----	----	---	---------------------

Example COMPARATIVE EXAMPLE 1

Parameter	Value	Unit	Context
Without	3.39	-	Material property

Example COMPARATIVE EXAMPLE 2

Parameter	Value	Unit	Context
Aliphatic	154	-	Material property
Without	2.6	-	Material property

PATENT US10266681

Example EXAMPLE 1

Parameter	Value	Unit	Context
Nylon	30	-	Material component
Fef	20	-	Material component
Dbu	3	-	Material component
Hexamethylenediamine	1.8	-	Material component

Example EXAMPLE 2

Parameter	Value	Unit	Context
Nylon	25	-	Material component

Example COMPARATIVE EXAMPLE 1

Parameter	Value	Unit	Context
Nylon	30	-	Material component
Product	40	-	Material property

Example EXAMPLE

Parameter	Value	Unit	Context
Product	40	-	Material property
Tensile	100	-	Material property

PATENT EP2463335

Example GENERAL PROCEDURE 1

Parameter	Value	Unit	Context
Per	30	-	Synthesis condition
Sol	70	-	Synthesis condition
Viscosity	100	-	Material property
Conversion	98	%	Synthesis condition

Methyl	80	-	Material component
Temperature	20	-	Synthesis condition
Because	10	-	Synthesis condition
Time	1	-	Synthesis condition
While	160	-	Synthesis condition
Nitrile	1	-	Material component
Wate	120	-	Synthesis condition
Acrylonitrile	45	-	Material component
2-vinylpyridine	2.2	-	Material component
Sodium	2	-	Material component
Emulsifier	20	-	Synthesis condition
Para-menthane	0.06	-	Material component
Iron	0.006	-	Material component
Initiator	80	-	Synthesis condition

Hydroxylamine	0.3	-	Material component
Potassium	0.06	-	Material component
Pressure	20	-	Synthesis condition
Solid Content	20	-	Material property
Distilled	995	-	Synthesis condition
Dilute	10	-	Synthesis condition
Under	50	-	Synthesis condition
325	0.5	-	Material component
Tetramethylthiuram	1.5	-	Material component
Ofn-cyclohexyl-2-benzothiazylsulfenamide	1.5	-	Material component
Vinyl	100	-	Material component
Ion-exchanged	150	-	Synthesis condition
Ammonium	0.3	-	Material component
T-dodecyhnercaptan	0.05	-	Material component

Styrene	24	-	Material component
T-dodecylmercaptan	0.05	-	Material component
Centrifuging	20	-	Synthesis condition
Not	10	-	Synthesis condition
Freeze-crushing	1	-	Synthesis condition
24	1	-	Synthesis condition
Iodine	1	-	Material component

PATENT EP3231826

Example EXAMPLE

Parameter	Value	Unit	Context
Until	70	-	Synthesis condition
Temperature	95	-	Synthesis condition
Conversion	95	%	Synthesis condition
Total	100	-	Synthesis condition
Maturation	1	-	Synthesis condition
Using	200	-	Synthesis condition
When	82	-	Synthesis condition

4. Cross-Patent Comparison

Comparing the ranges of key parameters across the patents reveals that: - Temperature ranges from 20 (EP2463335) to 128 (EP2492306) and 95 (EP3231826). - Viscosity is reported as 100 in both EP2492306 and EP2463335. - Conversion rates range from 95% (EP3231826) to 98% (EP2463335).

5. Key Technical Trends

Technical trends across the analyzed patents include: - The use of various initiators, emulsifiers, and cross-linking agents to optimize synthesis conditions. - A focus on achieving high conversion rates and specific material properties, such as viscosity and tensile strength. - The incorporation of different material components, including nylon, styrene, and vinylpyridine, to modify the properties of the resulting nitrile rubber compositions.

6. Most Relevant Patents

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year
EP2463335	Nitrile rubber composition, crosslinked nitrile rubber composition, crosslinked ...	Not disclosed	EP	2015
EP2492306	Cross-linkable nitrile rubber composition and manufacturing method for same	Not disclosed	EP	2014

US10266681	Cross-linkable nitrile rubber composition and cross-linked rubber product, and ...	Not disclosed	US	2019
EP3231826	Method for preparing acrylonitrile-butadiene-styrene graft copolymer, and ...	Not disclosed	EP	2020

7. Excluded / Rejected Patents

No patents were excluded or rejected, as all provided patents contained relevant synthesis or polymerization methods.

8. References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL
EP2492306	Cross-linkable nitrile rubber composition and manufacturing method for same	Not disclosed	EP	2014	https://patents.google.com

US10266681	Cross-linkable nitrile rubber composition and cross-linked rubber product, and ...	Not disclosed	US	2019	https://patents.google.com
EP2463335	Nitrile rubber composition, crosslinked nitrile rubber composition, crosslinked ...	Not disclosed	EP	2015	https://patents.google.com
EP3231826	Method for preparing acrylonitrile-butadiene-styrene graft copolymer, and ...	Not disclosed	EP	2020	https://patents.google.com